33V 6A 700KHz Fast-PWM Synchronous Step-Down Converter

DESCRIPTION

The MT3886 is a fully integrated high efficiency synchronous step-down converter which requires minimum number of external components. It offers very compact solution with up to 6A continuous output current over a wide input range. The MT3886 employs proprietary Constant On-Time (COT) control scheme providing superior transient response and maintaining constant switching frequency under the continuous conduction mode operation. The internal ramp compensation network allows stable operation with ultra-low equivalent series resistance (ESR) output ceramic capacitors without using external compensation network. An error amplifier in the control loop provides excellent line and load regulation. The MT3886 integrates extensive protection functions including: Input under-voltage lockout (UVLO) and over-voltage protection, thermal shutdown, output over-voltage protection (OVP), cycle by-cycle current limiting, and hiccup mode short-circuit protection. An open drain power good signal indicates the output within its regulation voltage range. The converter is available in a small QFN2X3_12L package.

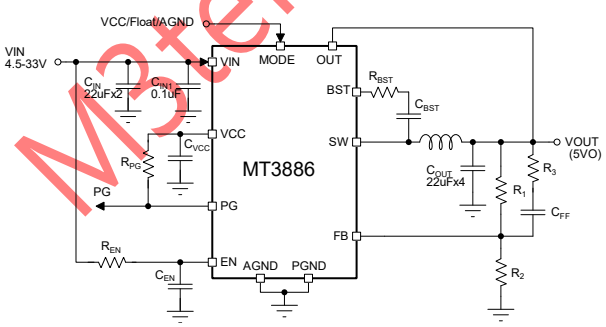
FEATURES

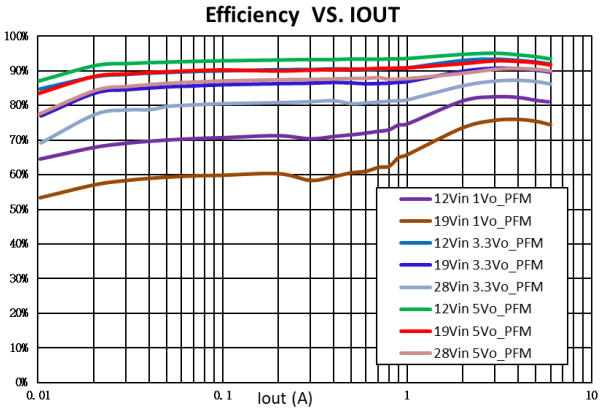
- Input Voltage Range from 4.5V to 33V
- Output Voltage Range from 0.6V to 33V
- Support Near 100% duty cycle Low Dropout Operation
- 1% 0.6V Feedback Voltage Accuracy
- 6A continuous output current
- Selectable 3 Operation Modes: PFM, Ultrasonic Mode (UAM) and Forced PWM Mode
- Stable operation with output low ESR ceramic capacitors
- 700kHz Switching Frequency
- Integrated 21mΩ and 10mΩ HS/LS-Switches
- Low quiescent current (PFM mode)
- Accurate enable threshold voltage
- Thermal Shutdown
- Short Circuit Hiccup Mode Protection:
- Small QFN2X3_12L package

APPLICATIONS

- USB Type-A or C and PD Dongle and Docking Station
- Tablet PC
- Networking Systems
- Flat Panel Television and Monitors
- Distributed Power Systems

TYPICAL APPLICATIONS





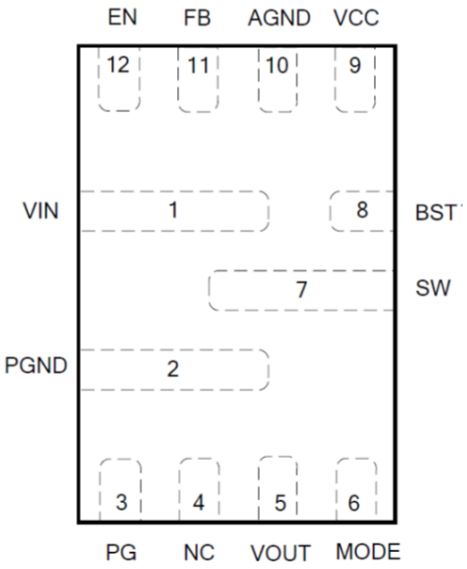
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ORDERING INFORMATION

Part No.	Marking	Temp. Range	Package	MOQ
MT3886MFFR	3886M YWWXX	-40°C ~+85°C	QFN2X3_12L	3,000/Reel

Note: Y: Year, WW: Week, XX: Control Code

PIN CONFIGURATION



QFN2x3_12L TOP VIEW

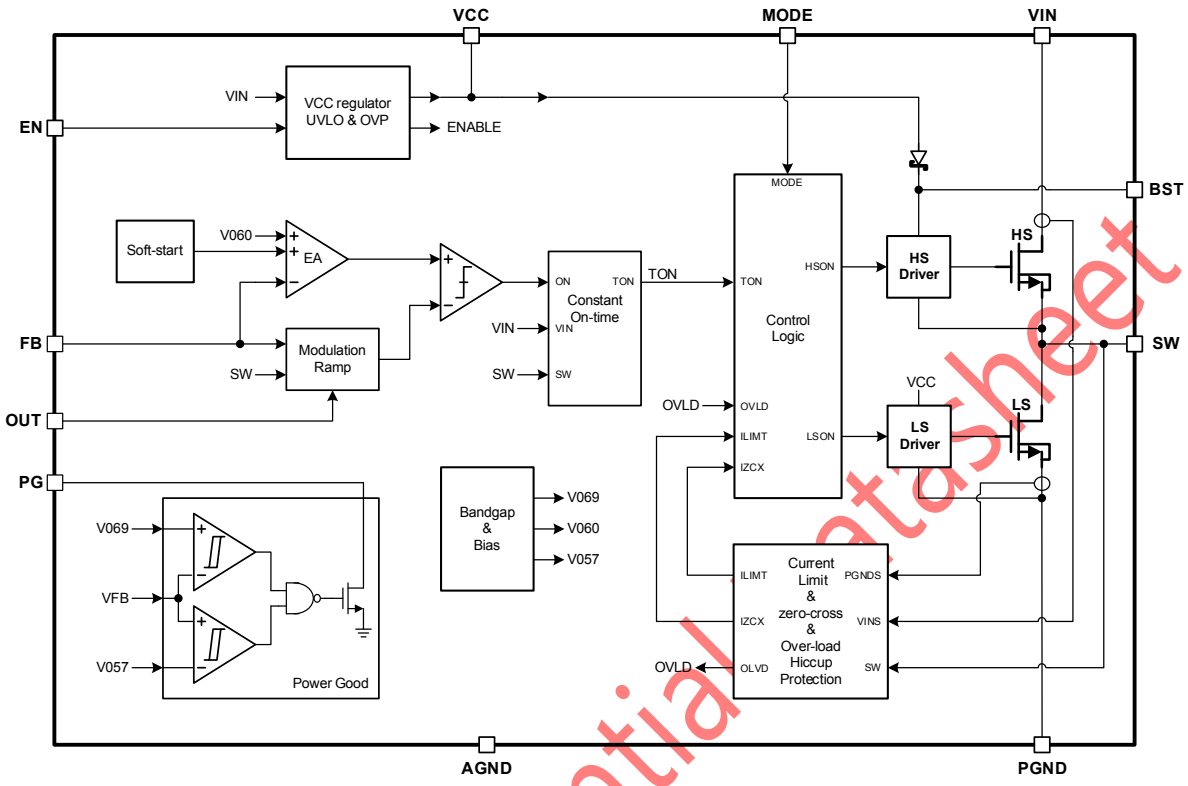
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PIN DESCRIPTION

QFN2x3_12L PIN No.	SYMBOL	DESCRIPTION
1	VIN	Input supply pin. Connect the input bypass capacitor close to VIN pin and PGND pin to minimize voltage spike and noises
2	PGND	Power ground pin
3	PG	Power good indication pin. PG is open drain output with internal pull down. PG is high-Z if the FB pin voltage is within 95% to 115% of regulation, otherwise PG is low-impedance to GND.
4	NC	No Connection. NC pin can be floated or connected to GND
5	OUT	Optional output sense pin. OUT can be connected to the output capacitor if the converter output is no higher than 5V. Leave OUT float or connected to AGND if the converter output is higher than 5V.
6	MODE	Operation mode selection pin. Connect MODE to Vcc to operate the MT3886 in the Forced-PWM (FPWM) mode; Leave MODE float (NC) to operate the MT3886 in Ultrasonic (UA) mode; with minimum switching frequency of 35kHz; Connect MODE to AGND to operate the MT3886 in Pulse-Frequency Modulation (PFM) mode.
7	SW	Switching node pin. Connected the external power inductor to SW
8	BST	Bootstrap pin for HS driver supply. Connect a resistor and capacitor between BST and SW pins
9	VCC	Power supply pin for internal control circuits. VCC is Internal 5V LDO output. Connect a min. 1uF capacitor from VCC to AGND.
10	AGND	Analog ground pin. Connect the ground of the FB divider resistor to AGND for best load regulation. Use Kelvin connection between AGND and PGND.
11	FB	Feedback voltage input pin. FB is regulated to 0.6V. Connecting FB to the tap point of the resistor divider from output to AGND sets the output voltage. A feedforward cap can be connected from the output to FB to improve the converter transient performance.
12	EN	Enable pin. MT3886 is in shut down mode when EN is pulled to below 0.4V. Pull EN above 1.26V to enable the converter. EN has internal 1MΩ pull down resistor to AGND.

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FUNCTIONAL BLOCK DIAGRAM



Absolute Maximum Ratings (Reference to AGND) (1)

Supply Voltage IN to GND..... +36V
EN to GND..... +36V
SW to GND..... -0.3V to V_{IN}+0.3V
SW Surge (30ns) to GND.....-3V to V_{IN}+3V
PGND to GND ..... -0.3V to +0.3V
BST to SW..... –0.3V to +6V
All Other Pins to GND..... –0.3V to +6V
Lead Temperature ..... +260°C
Junction Temperature ..... +150°C
Storage Temperature .....-65°C to +150°C

Recommend Operating Conditions (2)

Supply Voltage VIN .....+4.5V to +33V
Output Voltage VOUT ..... 0.6V to V_{IN}
Operating Junction Temp. (TJ)..... -40°C to +125°C

Thermal Information (3) (4)

Maximum Power Dissipation(T_A=+25°C)
QFN2X3_12L..... 2.22W
Thermal Resistance (θ_{JA}) QFN2X3_12L.....56.2°C/W
Thermal Resistance (θ_{JC}) QFN2X3_12L.....23.4°C/W

Note(1): Stress exceeding those listed “Absolute Maximum Ratings” may damage the device.
Note(2): The device is not guaranteed to function outside of the recommended operating conditions.
Note(3): Measured on JESD51-7, 4-Layer PCB.
Note(4):The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by $P_{D_MAX}= (T_{J_MAX}-T_A)/\theta_{JA}$. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

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ELECTRICAL CHARACTERISTICS

T_A = +25°C, unless otherwise noted. Typical values are at V_{IN} = 12 V, V_{OUT} =V_{EN}= 5.0V.

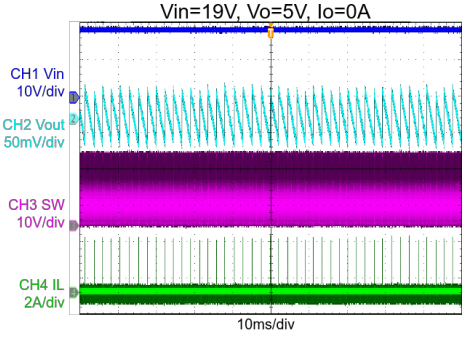
PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input Operating Voltage Range	V_{IN}		4.5		33	V
Input Under Voltage Lockout Threshold Rising	V_{INUVLO}		4.00	4.2	4.40	V
Input Under Voltage Lockout Threshold Hysteresis	V_{UVLO_HYS}			300		mV
Input Over Voltage Protection Threshold Rising	V_{INOVP}		35.5	36.7	37.9	V
Input Over Voltage Protection Threshold Hysteresis	V_{OVP_HYS}			2		V
Input Shutdown Current	I_{SD}	V_{EN}=0V		5	9	uA
Input Quiescent Current	I_Q	V_{EN}=2V, VFB=0.63V, MODE=GND (PFM mode)	70	100	130	uA
Feedback Reference Voltage	V_{REF}		598	604	610	mV
Feedback Input Current	I_{FB}	V_{FB}=0.604V		1	50	nA
Switching Frequency	F_{SW}		560	700	840	kHz
Maximum Duty Cycle	D_{MAX}	V_{OUT}=5V, V_{IN}=5V		99.5		%
Minimum Off Time	T_{OFF_MIN}		130	160	190	nS
Soft Start Time	T_{SS}		1.4	1.8	2.2	ms
High-side Switch On Resistance	R_{ONHS}			21		mΩ
High-side Switch Leakage Current	I_{LKGHS}	V_{IN}=33V, V_{EN}=0V, V_{SW}=0V			2	uA
High-Side Switch Peak Current Limit	I_{PK}		9	11	13	A
Low-side Switch On Resistance	R_{ONLS}			10		mΩ
Low-side Switch Leakage Current	I_{LKGLS}	V_{IN}=33V, V_{EN}=0V, V_{SW}=33V			2	uA
Low-Side Switch Valley Current Limit	I_{VLY}		8	10	12	A
Low-side Switch Zero-Cross current	I_{ZX}	PFM /UA mode		100		mA
FPWM MODE Threshold	V_{MODE_H}		1.4	1.8	2.2	V
PFM MODE Threshold	V_{MODE_L}		0.3	0.6	0.9	V
MODE pull-down current	$I_{MODE\ PD}$		0.5	1	1.8	uA
EN On Voltage	V_{ENON}		1.22	1.26	1.30	V
EN Off Voltage	V_{ENOFF}		1.12	1.15	1.20	V
EN Pull-Down Resistor	R_{EN}	V_{EN}=2V	0.6	1	1.4	MΩ
VCC Regulator	V_{CC}		5.0	5.2	5.4	V
VCC Load Regulation		I_{VCC}=20mA		1		%
PG FB UV Falling Threshold (Fault)	PG_{UV_LO}		80	85	90	%
PG FB OV Rising Threshold (Fault)	PG_{UV_HI}		110	115	120	%
Power Good Fault Delay	PG_{TD}		150	200	250	us
Power Good Sink Current Capability	V_{PG}	PG pin sink 3mA	0.2	0.3	0.45	V
Power Good Leakage Current	I_{LKGPG}	V_{PG}=5V			10	nA
Thermal Shutdown (GBD)	T_{OTP}			160		°C
Thermal Shutdown Hysteresis	T_{OTP_HYS}			30		°C

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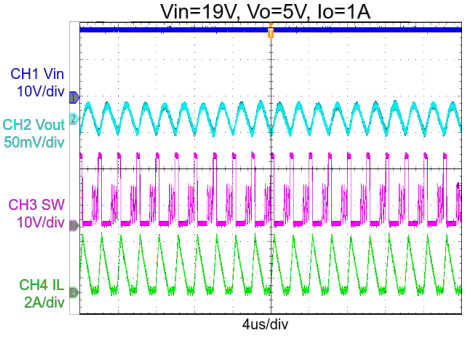
TYPICAL PERFORMANCE CHARACTERISTICS

C_{IN}=2x22uF, C_{OUT}=4x22uF, L=1.5uH, T_A=+25°C.

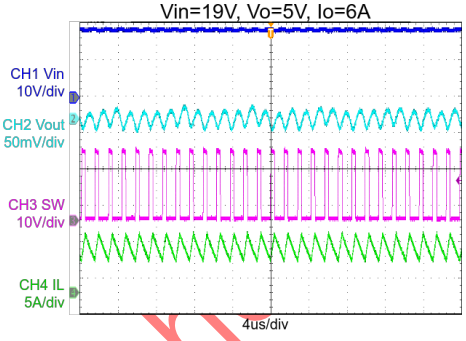
Steady State Test
Vin=19V, Vo=5V, Io=0A



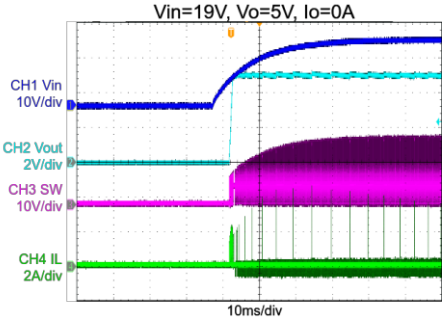
Steady State Test
Vin=19V, Vo=5V, Io=1A



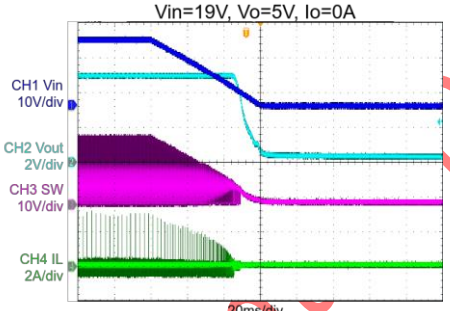
Steady State Test
Vin=19V, Vo=5V, Io=6A



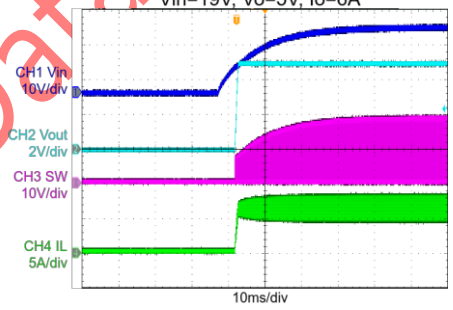
Vin Power On
Vin=19V, Vo=5V, Io=0A



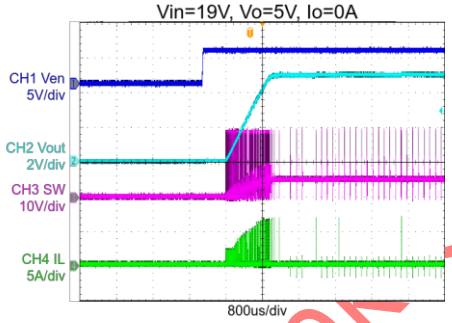
Vin Power Off
Vin=19V, Vo=5V, Io=0A



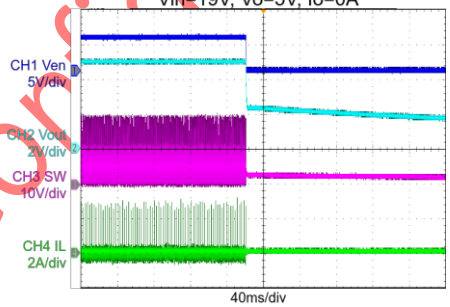
Vin Power On
Vin=19V, Vo=5V, Io=6A



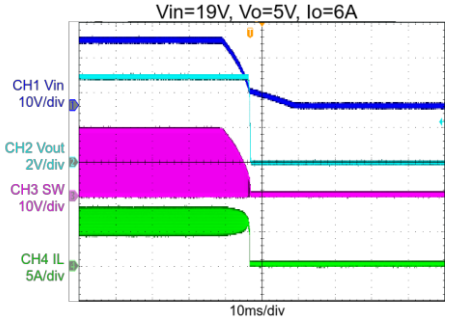
EN Power On
Vin=19V, Vo=5V, Io=0A



EN Power Off
Vin=19V, Vo=5V, Io=0A



Vin Power Off
Vin=19V, Vo=5V, Io=6A



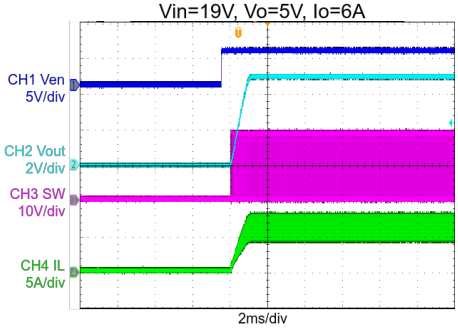
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TYPICAL PERFORMANCE CHARACTERISTICS

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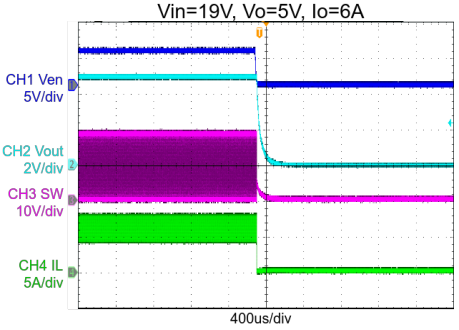
EN Power On

Vin=19V, Vo=5V, Io=6A



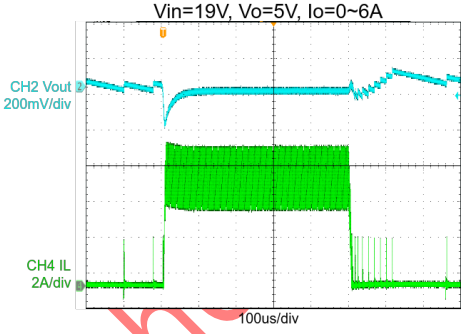
EN Power Off

Vin=19V, Vo=5V, Io=6A



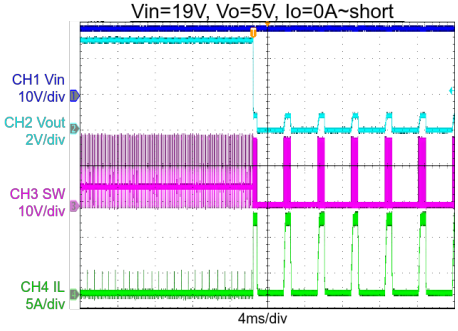
Load Transient Response

Vin=19V, Vo=5V, Io=0~6A



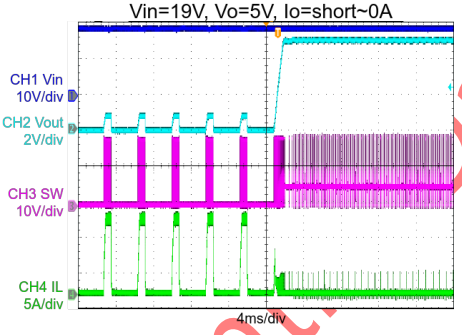
Short Protection

Vin=19V, Vo=5V, Io=0A~short



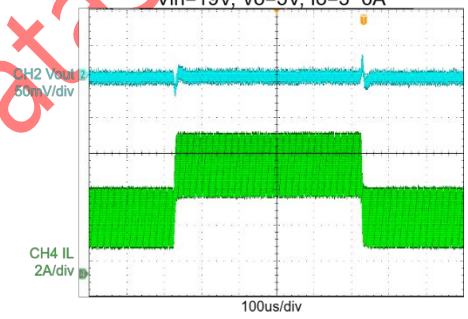
Short Protection

Vin=19V, Vo=5V, Io=short~0A

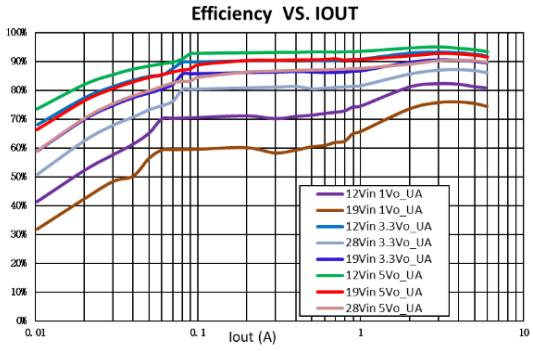


Load Transient Response

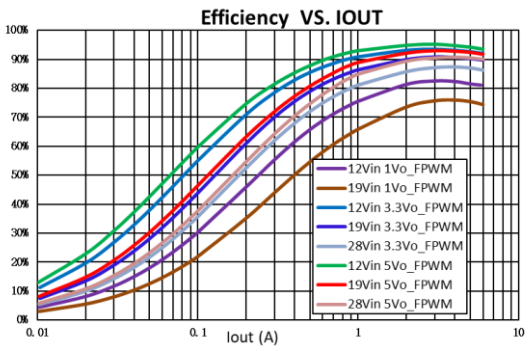
Vin=19V, Vo=5V, Io=3~6A



Efficiency VS. IOUT



Efficiency VS. IOUT



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OPERATION

The MT3886 is a fully integrated synchronous step-down converter employing an advanced constant on-time (COT) control scheme to achieve superior transient performance. With the compensation ramp of the control loop set by the feed-forward cap, MT3886 has adjustable loop compensation hence can be optimized to have the best transient response for a wide range of applications.

Fast-PWM Constant On-Time Control

The MT3886 employs M3Tek's proprietary constant on-time control (Fast-PWM) to achieve superior transient response. The controller compares the feedback voltage V_{FB} with an internal reference voltage V_{REF}. When V_{FB} drops below V_{REF} level, the high-side (HS) switch is immediately turned on for a pre-determined time (on-time) to ramp up the inductor current. Once the on-time ends, the HS switch is turned off and the low-side (LS) switch is turned on to ramp down the inductor current. The LS switch remains on until the inductor current reaches zero (PFM or UA mode) or the next time V_{FB} drops below V_{REF}, where the LS switch is turned off and HS switch is turned on again. In steady state operation, the controller modulates the turn-off time point of the LS switch so that the inductor current always ramps up and ramps down with the exact same amplitude and the cycle repeats itself.

The MT3886 uses a proprietary algorithm to calculate the on-time based on input voltage, output voltage and load current to achieve a nearly constant switching frequency (F_{SW}) of 700kHz over the entire continuous conduction load current range. The on-time can be estimated from:

$$T_{ON} = \frac{V_{OUT}}{V_{IN} * F_{SW}} = \frac{V_{OUT}}{V_{IN}} * 1.428\mu s$$

Due to its immediate response on FB voltage drop and simplified loop compensation, the MT3886 offers superior transient response compared to the traditional fixed frequency PWM control converters.

MODE SELECTION

The MT3886 comes with a MODE pin that allows the user to select one of three operation modes for the converter; Pull MODE to higher than 2.4V to operate the MT3886 in FPWM (Forced-PWM) mode for constant switching frequency over full load range; Float MODE to operate the MT3886 in UA (Ultra-sonic) mode to avoid audible noise while maintain light load efficiency; Pull MODE to below 0.3V to operate the MT3886 in PFM (Pulse-Frequency Modulation) mode for maximum light load efficiency.

PFM/UA/FPWM Mode Operation

In PFM/UA mode operation, the MT3886 keeps both HS and LS off at light load to maintain high efficiency of the converter. As load current decreases from medium or heavy load, the MT3886 will transition from 700kHz continuous conduction mode (CCM) to discontinuous conduction mode (DCM) where the inductor current drops to zero between pulses and switching frequency starts to decrease (PFM mode). As soon as the inductor current reaches zero, the MT3886 turns off low-side (LS) switch and then the SW node becomes high-Z state until the next

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cycle when the high-side (HS) switch is turned on again. The HS switch on-time in DCM mode is the same as that in CCM mode, hence each switching pulse in DCM mode provides the same amount of energy to the converter output. As load current continues to decrease, the switching pulses also become proportionally less frequent. When the switching frequency drops to a low enough level, the device enters sleep mode to cut down most of its quiescent current to further reduce power consumption.

The critical load current I_{CRIT} at the boundary of CCM mode and DCM mode is determined by the inductor ripple current, which depends on the inductor value, input voltage and output voltage. I_{CRIT} can be estimated from:

$$I_{CRIT} = \frac{1}{2} * \frac{(V_{IN} - V_{OUT}) * V_{OUT}}{L * F_{SW} * V_{IN}}$$

When the load current is below this I_{CRIT} level, the MT3886 will be operating in PFM/UA mode.

In PFM/UA mode, the SW node is at output voltage level when both HS and LS switches are turned off. To maintain the HS driver floating supply voltage on BST cap, the MT3886 refreshes BST cap voltage by turning on LS switch for about 130ns at a fixed time duration when there is no switching pulse on SW. For PFM mode, this time duration is about 150us. For UA mode, this time duration is about 30us which is above the audio frequency range to avoid audible noises.

FPWM Mode Operation

In FPWM mode operation, the MT3886 switches the high-side (HS) and low-side (LS) alternatively at fixed 700kHz to maintain a constant frequency spectrum over the full load current range. In steady state, the inductor current ramps up to a peak point when the HS is turned on, and ramps down to a valley point when the LS is turned on, while the average level of the peak and valley point equals the load current. Thus, at no load or very light load condition, the valley point of the inductor current is negative and this is the normal behavior of the FPWM mode operation.

The FPWM mode is suitable for applications requiring tight control of the switching frequency spectrum or small output voltage ripple, at the cost of significantly lower efficiency at light load.

100% Duty Cycle Low Dropout Operation

When the input voltage drops approaching the output voltage, the MT3886 will extend the high-side (HS) switch on-time accordingly or beyond the normal switching period to satisfy the duty cycle requirement to regulate the output voltage. If the input further drops to equal or lower than the target output level, the MT3886 will extend the HS switch on-time to the maximum duration of 30us to achieve an effective 100% duty cycle operation. When the maximum on-time times out, HS switch is turned off and low-side (LS) switch is turned on for about 130ns to refresh the BST cap voltage. Once this refresh pulse finishes, the HS switch is immediately turned on to for another maximum on-time of 30us. With periodically refreshing the BST cap voltage, the HS switch driver floating supply is maintained at sufficient voltage level in 100% operation mode.

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Input Under-Voltage Lockout (UVLO) and Over-Voltage Protection (OVP)

The MT3886 has built-in input under-voltage lockout (UVLO) and over-voltage protection (OVP). When input voltage (V_{IN}) is lower than the UVLO threshold voltage or higher than the OVP threshold voltage, MT3886 will stop operating to prevent malfunction or damage. The UVLO rising threshold voltage of 4.2V with hysteresis of 300mV, and the OVP rising threshold voltage is 36.7V with hysteresis of 2.0V.

Enable Control

When the input supply (V_{IN}) is in valid range, i.e., above UVLO threshold and below OVP threshold, MT3886 can be enabled or disabled by the enable pin (EN). When EN pin voltage is below 400mV, the MT3886 is in shutdown mode where it only consumes 5uA current.

MT3886 offers an accurate enable threshold for the EN pin, which is 1.26V rising and 1.15V falling. MT3886 is enabled when the EN pin voltage is above 1.26V and disabled when the EN pin voltage is below 1.15V. The accurate EN pin threshold provides the option to program input startup threshold (higher than the built-in UVLO) with resistor divider from input to ground. The programmed input startup threshold can be calculated with the following equation:

$$V_{IN-START} = 1.26V * \frac{R_{TOP} + R_{BOT}//1M\Omega}{R_{BOT}//1M\Omega}$$

Where the 1MΩ is the internal pull-down resistor from EN pin to ground.

Soft Start

The MT3886 has an internal built-in soft-start time of 1.8ms. During the soft start period, output voltage is ramped up linearly to the regulation voltage, independent of the load current level and output capacitor value.

Power Good Indication (PG)

The MT3886 has an open-drain power good (PG) indication pin. The PG pin is the drain of an internal pull down MOSFET and should be connected to VCC or another voltage source (<5.5V) with a pull-up resistor. The recommended resistor value is 10kΩ to 500kΩ.

When the feedback pin voltage (V_{FB}) rises above 95% of regulation or falls below 105% of regulation, the PG pin is pulled high by the external resistor. When V_{FB} drops below 85% of regulation or rises above 115% of regulation, the PG is pulled low by the internal pull down MOSFET. To prevent glitching both the upper and lower PG trip thresholds come with approximately 10% hysteresis.

Over-Current Protection (OCP) and Short Circuit Fault Protection

The MT3886 has built-in cycle-by-cycle current limit protection to prevent inductor current from running away in any fault conditions. The MT3886 continuously monitors the inductor peak and valley current during its operation. Once the peak current exceeds the peak limit level, MT3886 immediately turns off the high-side (HS) switch and turns on

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low-side (LS) switch and wait for the inductor current to drop down to a pre-determined level before the HS switch can be turned on again. If this current limit condition is repeated for a sustained long period of time, MT3886 will consider the output is over-load or short circuit. Either way, MT3886 will enter hiccup mode, where it stops switching for a pre-determined period of time before automatically re-try to start up again. It always starts up with soft-start to limit inrush current and avoid output over-shoot.

Thermal Shutdown

The MT3886 has internal thermal shutdown to prevent the device from overheating. When the device junction temperature reaches the threshold temperature of 160°C, the device stops switching and waits for the junction to cool down. When the junction temperature drops to 130°C, the device resumes operation with soft-start.

APPLICATION INFORMATION

Setting the Output Voltage

External feedback resistors are used to set the output voltage. 1% resistors are recommended to maintain output voltage accuracy. The feedback network is shown Figure 1 below. The top feedback resistor R_1 has some impact on the loop stability, R_1 recommended range is 20KΩ~100KΩ. Choose R_1 around 50KΩ as a start point. For any chosen R_1, the bottom feedback resistor R_2 can be calculated as:

$$R_2 = \frac{R_1}{\frac{V_{OUT}}{0.600} - 1}$$

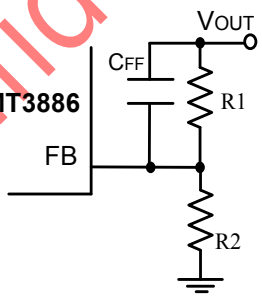


Figure 1: Feedback Network

Inductor

The inductor is necessary to supply constant current to the output load while being driven by the switched input voltage. A larger-value inductor will result in less ripple current that will result in lower output ripple voltage. However, a larger-value inductor will have a larger physical footprint, higher series resistance, and/or lower saturation current. A good rule for determining the inductance value is to design the peak-to-peak ripple current in the inductor to be in the range of 30% to 50% of the maximum output current, and that the peak inductor current is below the maximum switch current limit. The inductance value can be calculated by:

$$L = \frac{V_{OUT}}{F_{SW} \times \Delta I_L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

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Where ΔI_L is the peak-to-peak inductor ripple current and switching frequency F_{SW} is typical 700KHz.

To avoid overheating and poor efficiency, an inductor must be chosen with an RMS current rating that is greater than the maximum expected output load of the application. In addition, the saturation current (typically labeled as ISAT) rating of the inductor must be higher than the maximum load current plus 1/2 of in inductor ripple current.

The peak inductor current can be calculated by:

$$I_{LP} = I_{OUT_MAX} + \frac{V_{OUT}}{2F_{SW} \times L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

The current flowing through the inductor is the inductor ripple current plus the output current. During power-up, faults or transient load conditions, the inductor current can increase above the peak inductor current level which is calculated above. In transient conditions, the inductor current can increase up to the switch current limit of the device. For this reason, the most conservative approach is to specify an inductor with a saturation current rating equal to or greater than the switch current limit rather than the peak inductor current.

Input Capacitor

The input current to the step-down converter is discontinuous and therefore requires a capacitor to supply the AC current to the step-down converter while maintaining the DC input voltage. The MT3886 requires a high-quality ceramic, type X5R or X7R, input decoupling capacitor of at least 44µF of effective capacitance should be placed as close to the VIN pins as possible. In some applications, additional bulk capacitance may also be required for the VIN input. The effective capacitance includes any DC bias and temperature de-rate effects. The voltage rating of the input capacitor must be greater than the maximum input voltage.

The capacitors must also have a ripple current rating greater than the maximum input ripple current of the converter. The input ripple current can be estimated as follows:

$$I_{CIN} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})}$$

The worst-case condition occurs at V_{IN}= 2*V_{OUT}, where:

$$I_{CIN} = \frac{I_{OUT}}{2}$$

For simplification, choose the input capacitor with an RMS current rating greater than half of the maximum load current. The input capacitance value determines the input voltage ripple of the converter. If there is an input voltage ripple requirement in the system, choose the input capacitor that meets the specification. The input voltage ripple can be estimated as follows:

$$\Delta V_{IN} = \frac{I_{OUT}}{F_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Under worst-case conditions where V_{IN} = 2*V_{OUT}:

$$\Delta V_{IN} = \frac{1}{4} \times \frac{I_{OUT}}{F_{SW} \times C_{IN}}$$

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Output Capacitor

The output capacitor has two essential functions. Along with the inductor, it filters the square wave generated by the MT3886 to produce the DC output. In this role it determines the output ripple, thus low impedance at the switching frequency is important. The second function is to store energy in order to satisfy transient loads and stabilize the MT3886's control loop. X5R or X7R types ceramic capacitors have very low equivalent series resistance (ESR) and provide low output ripple and good transient response. Transient performance can be improved with a higher value output capacitor and the addition of a feed-forward capacitor C_{FF} placed between V_{OUT} and FB. Increasing the output capacitance will also decrease the output voltage ripple. A lower value of output capacitor can be used to save space and cost but transient performance will suffer and may cause loop instability. When choosing a capacitor, special attention should be given to the data sheet to calculate the effective capacitance under the relevant operating conditions of voltage bias and temperature. A physically larger capacitor or one with a higher voltage rating may be required.

The output voltage ripple can be estimated with Equation below:

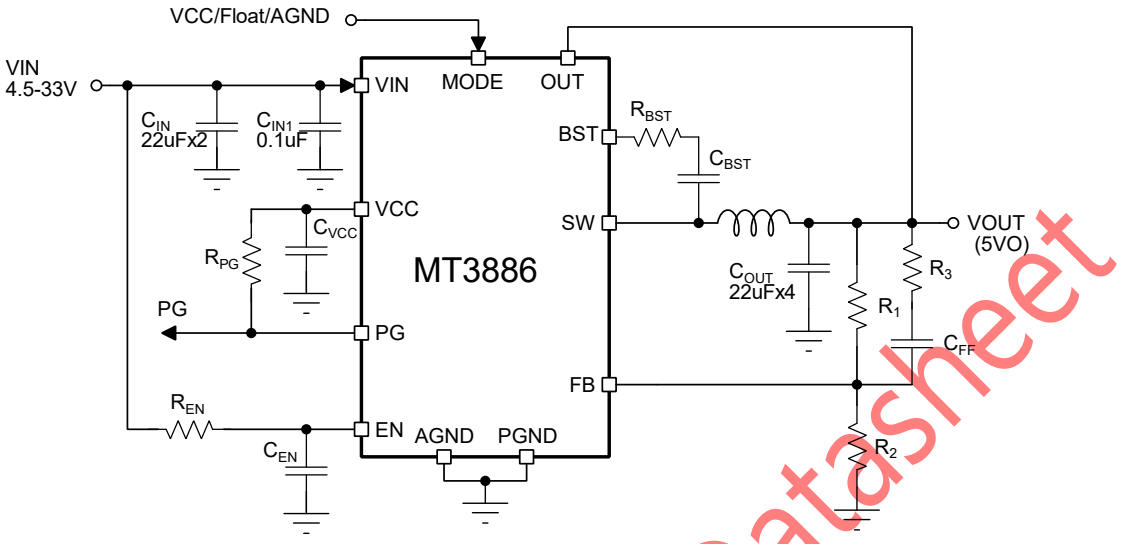
$$\Delta V_{OUT} = \frac{V_{OUT}}{F_{SW} \times L} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \times \left(R_{ESR} + \frac{1}{8 \times F_{SW} \times C_{OUT}}\right)$$

With ceramic capacitors, the impedance at the switching frequency is dominated by the capacitance. The output voltage ripple is mainly caused by the capacitance.

The value of a ceramic capacitor varies significantly over temperature and the amount of DC bias applied to the capacitor. The capacitance variations due to temperature can be minimized by selecting a dielectric material that is stable over temperature. X5R and X7R ceramic dielectrics are usually selected for power regulator capacitors because they have a high capacitance to volume ratio and are fairly stable over temperature. The output capacitor must also be selected with the DC bias taken into account. The capacitance value of a capacitor decreases as the DC bias across a capacitor increases. For USB Type-C/PD application design, four 22μF ceramic capacitors with 35V voltage rating or higher in parallel are recommended to support the maximum output voltage 28V.

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Application Schematic 1: OUT Pin Connect To VOUT

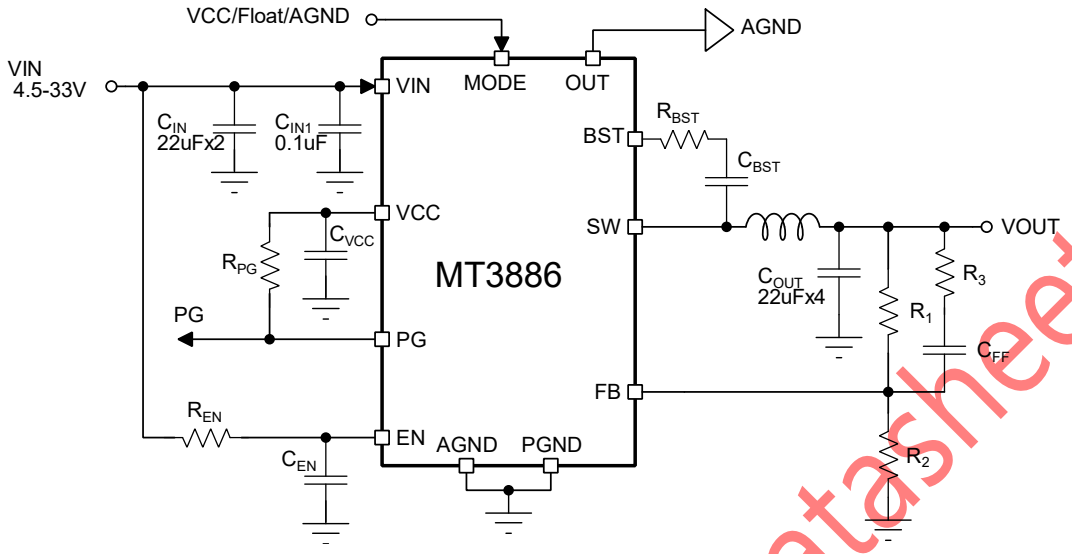


EVB BOM List

Qty	Ref	Value		Description	Package
2	C_{IN}	22μF		Ceramic Capacitor, 50V, X5R	1206
4	C_{OUT}	22uF		Ceramic Capacitor, 35V, X5R	0805
1	C_{IN1}	0.1μF		Ceramic Capacitor, 50V, X5R	0603
2	C_{EN}, C_{BST}	220nF		Ceramic Capacitor, 50V, X5R	0603
1	C_{VCC}	1μF		Ceramic Capacitor, 50V, X5R	0603
1	L	Vout=5V	1.5μH	Wurth Electronics 7443552150, DCR=5.3 mΩ	SMD
		Vout=3.3V	1.5μH	Wurth Electronics 7443552150, DCR=5.3 mΩ	
		Vout=1.0V	0.68μH	Wurth Electronics 744311068, DCR=3.1mΩ	
1	R1	49.9KΩ		Resistor, ±1%	0603
1	R2	Vout=5V	6.8KΩ	Resistor, ±1%	0603
		Vout=3.3V	11KΩ		
		Vout=1.0V	75KΩ		
1	C_{FF}	Vout=5V	68pF	Ceramic Capacitor, 50V, X5R	0603
		Vout=3.3V	68pF		
		Vout=1.0V	47pF		
1	R3	1KΩ		Resistor, ±1%	0603
1	R_{BST}	10Ω		Resistor, ±1%	0603
2	R_{PG}, R_{EN}	100KΩ		Resistor, ±1%	0603
1	Power IC	MT3886		Step-Down DC/DC Converter	QFN2x3_12L

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Application Schematic 2: OUT Pin Connect To AGND For Type-C



EVB BOM List

Qty	Ref	Value		Description	Package
2	C_{IN}	22μF		Ceramic Capacitor, 50V, X5R	1206
4	C_{OUT}	22uF		Ceramic Capacitor, 35V, X5R	0805
1	C_{IN1}	0.1μF		Ceramic Capacitor, 50V, X5R	0603
2	C_{EN}, C_{BST}	220nF		Ceramic Capacitor, 50V, X5R	0603
1	C_{VCC}	1μF		Ceramic Capacitor, 50V, X5R	0603
1	C_{FF}	33pF		Ceramic Capacitor, 50V, X5R	0603
1	L	Vout=5V Vout=9V Vout=12V Vout=20V	6.8μH	Wurth Electronics 74439346068, DCR=17.6mΩ	SMD
1	R1	150KΩ		Resistor, ±1%	0603
1	R2	Vout=5V Vout=9V Vout=12V Vout=20V	20.5KΩ 10.7KΩ 7.87KΩ 4.64 KΩ	Resistor, ±1%	0603
1	R3	1KΩ		Resistor, ±1%	0603
1	R_{BST}	10Ω		Resistor, ±1%	0603
2	R_{PG}, R_{EN}	100KΩ		Resistor, ±1%	0603
1	Power IC	MT3886		Step-Down DC/DC Converter	QFN2x3_12L

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PCB Layout Recommendation

The physical design of the PCB is the final stage in the design of power converter. If designed improperly, the PCB could radiate excessive EMI and contribute instability to the power converter. Therefore, following the PCB layout guidelines below can ensure better performance of MT3886.

(1) The loop (Vin-SW-L-Cout-GND) indicates a high current path. The traces within the loop should be kept as wide and short as possible to reduce parasitic inductance and high-frequency loop area. It is also good for efficiency improvement.

(2) Input capacitor as close as possible to the IC Pins (Vin and GND) and the input loop area should be as small as possible to reduce parasitic inductance, input voltage spike and noise emission.

(3) Keep the switching node SW short and away from the feedback network.

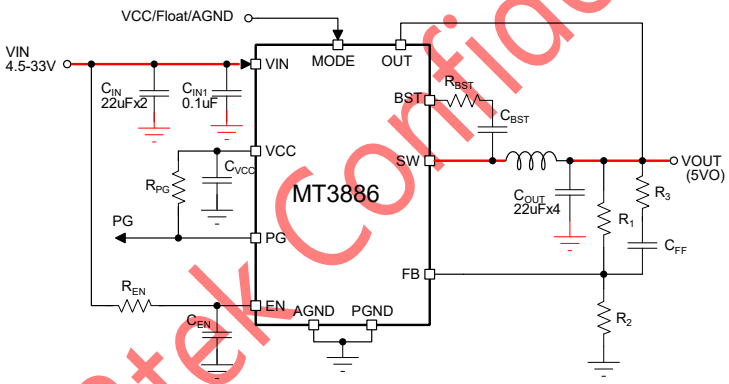
(4) The external feedback resistors should be placed next to the FB pin. Make sure that there is no via on the FB trace.

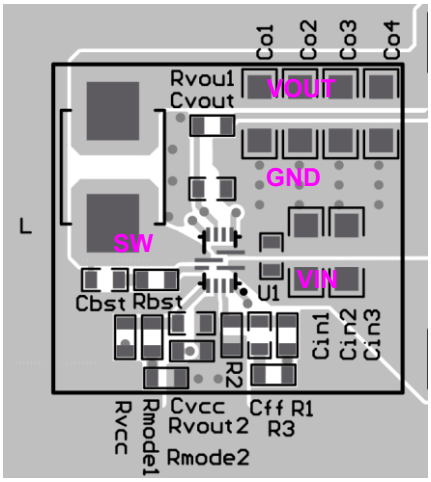
(5) Keep the BST voltage path (BST, R_{BST}, C_{BST} and SW) as short as possible.

(6) Keep the IN and PGND pads connected with large copper and use at least two layers for IN and PGND trace to achieve better thermal performance.

(7) Four-layer layout is strongly recommended to achieve better thermal performance.

Note: Please refer to the PCB Layout Application Notes for more details.

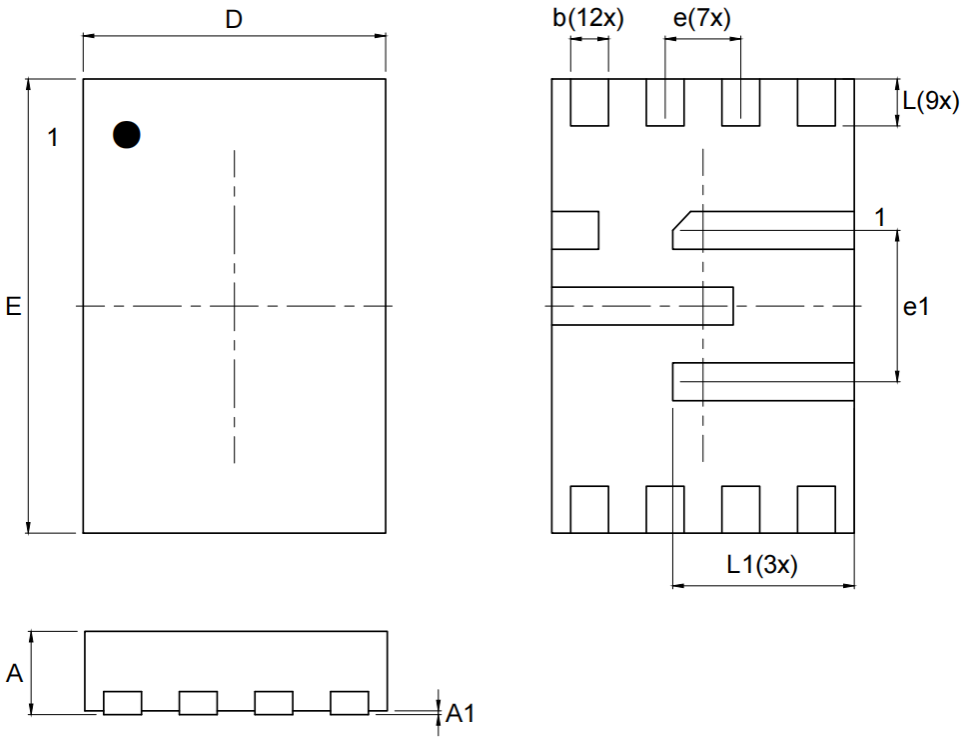




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PACKAGING INFORMATION

QFN 2x3 _12L

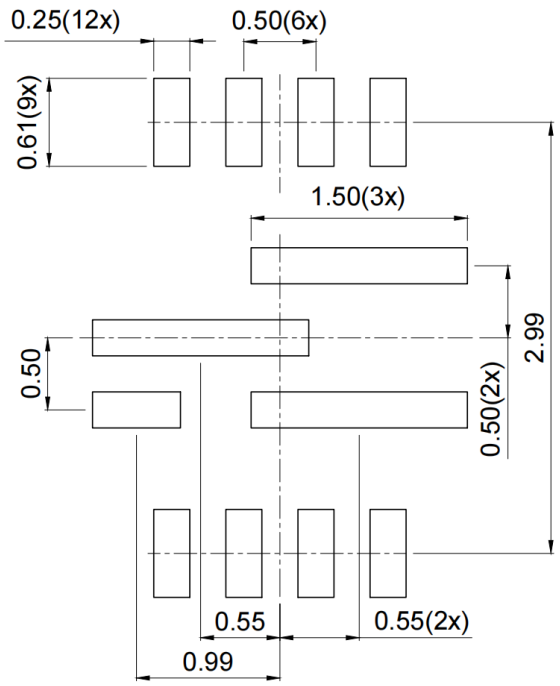


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.50	0.60	0.020	0.024
A1	0.00	0.05	0.000	0.002
b	0.20	0.30	0.008	0.012
D	1.90	2.10	0.075	0.083
E	2.90	3.10	0.114	0.122
e	0.50 BSC		0.020 BSC	
e1	1.00 BSC		0.039 BSC	
L	0.26	0.36	0.010	0.014
L1	1.15	1.25	0.045	0.049

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LAND PATTERN INFORMATION

QFN 2x3 _12L



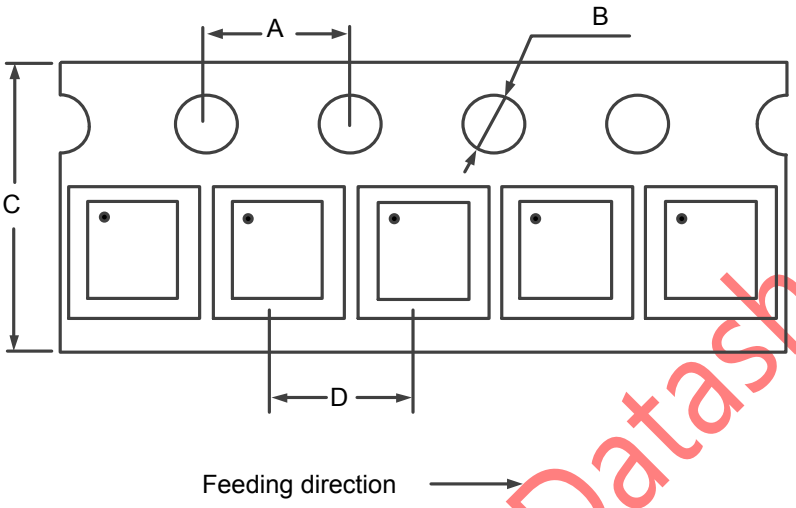
Note:

- Dimensions in mm.
- For reference only.

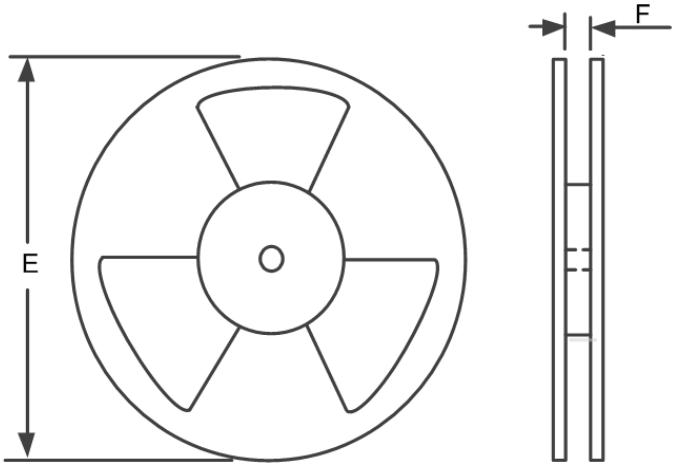
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CARRIER TAPE & REEL DIMENSIONS

1. Orientation / Carrier Tape Information:



2. Reel Information:



3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
QFN 2x3 _12L	4.0 mm	1.5 mm	8.0 mm	4.0 mm	7 inches	9.0 mm	3,000

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REFLOW PROFILE

Classification Of IR Reflow Profile

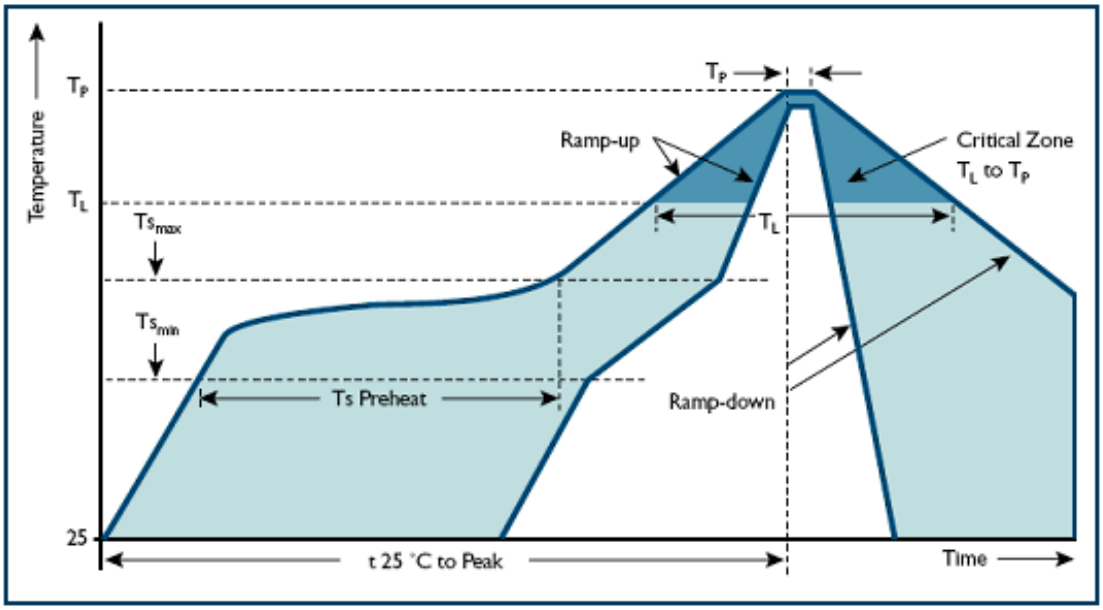
Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ts_{min} to Tp)	1~2°C/second, 3°C/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time($ts_{min}$to $tsts_{max}$)	 150°C 200°C 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217°C 60~150 seconds
Peak Temperature(Tp)	See Classification Temp intable1
Time within 5°Cof actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6°C/second max.
Time 25°C to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature(T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p)is defined as a supplier minimum and a user maximum.

Table 1. Pb-freeProcess –Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6mm	260°C	260°C	260°C
1.6mm–2.5mm	260°C	250°C	245°C
2.5mm	250°C	245°C	245°C

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.



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